

RESEARCH ARTICLE

Genome-wide characterization and expression pattern of auxin response factor (ARF) gene family in soybean and common bean

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Abstract Auxin response factors (ARFs) encode one of the most abundant groups of auxin-mediated response transcription factors in higher plants and play a major role in various biological processes. The success of whole genome sequencing allows for comprehensive phylogenetic analysis of ARF genes in plants. Here, we identified 80 ARF genes belonging to five different groups in legume species, soybean (*Glycine max*, 55) and common bean (*Phaseolus vulgaris*, 25), based on phylogenetic analysis and supported by motif analysis. There is no member ARF of common bean in group V which contained two members from soybean, GmARF12 and GmARF55. The duplication events among two species were also observed by using *Ks* values. A majority of the ARF genes (96.1 %; 38 GmARFs and 56.0 %, 14 PvARFs) were segmentally duplicated while only two and one of ARF *G. max* and *P. vulgaris*

genes were tandemly duplicated, respectively. In addition, expression profiling indicated that ARF genes in soybean and common bean perform various functions in plant growth and development, excluding flowering development. Furthermore, diversity of *cis*-elements in promoter regions also revealed differential expression of ARF proteins.

Keywords Auxin response factor · *Cis*-elements · Duplication events · *Glycine max* · *Phaseolus vulgaris* · Phylogenetic analysis

Introduction

Regulation of various aspects of plant growth and development are attributed to the phytohormone auxin, which is characterized by indole-3-acetic acid (IAA). Increasing evidence suggests that auxin participates in regulation of responses to environmental stimuli such as apical dominance, vascular differentiation, tropic responses, lateral root formation, shoot elongation, and embryo partitioning. In angiosperms, auxin partially controls plant developmental processes either by induction or repression of related genes (Zhao 2010). At the molecular level, auxin rapidly induces gene expression. The role of auxin is noticeable in the enhancement of transcription of numerous early gene classes, such as small auxin up RNA, Gretchen Hagen3, Aux/IAA, and auxin responsive factors (ARF) (Calderon Villalobos et al. 2012). Expression profiling has revealed that many auxin-responsive genes are responsive to various abiotic stressors. Additionally, *Arabidopsis* and rice mutants with mutations in ARF genes have resulted in leaf senescence and floral organ abscission (ARF12), differential cell division and elongation (ARF719),

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gynoecium and floral meristem patterning (ARF3), abnormal vascular strands and embryo axis (ARF5), impaired hypocotyl response to blue light and auxin stimuli (ARF7), infertile closed buds with short petals, short stamen filaments, un-dehiscent anthers (ARF6, 8), low growth, poor vigor, curled leaves, altered organ size (OsARF1), iron homeostasis (OsARF12), and iron and phosphate utilization (OsARF16) (Attia et al. 2009; Ellis et al. 2005; Krogan et al. 2012; Kumar et al. 2011; Narise et al. 2010; Qi et al. 2012; Schruoff et al. 2006; Shen 2013).

The regulation of plant physiological processes is controlled by modulating the interaction of auxin response elements (AuxREs) of the affected genes with transcription factors. Auxin gene expression is mediated by ARF and Aux/IAA proteins. ARF proteins bear three characteristic domains, i.e. a DNA-binding domain responsible for recognition of AuxREs present in promoters, a middle domain having transcriptional activation/repression ability, and a C-terminal domain (hosting the III and IV motifs) with an ability to mediate dimerization (Hagen et al. 2002; Ulmasov et al. 1999a, b). Transcriptional activation/repression activity of an ARF protein solely depends upon the amino acid sequence present in the middle domain (Tiwari et al. 2003; Ulmasov et al. 1999a). ARF middle domains bearing the transcriptional activation activity are rich in leucine, serine, and glutamine, on the contrary, serine, leucine, and proline and/or glycine are more prevalent in domains responsible for repression activity. The SCF^{TIR1} ubiquitin ligase complex ubiquitinates and degrades the Aux/IAAs, whereas auxin facilitates their interaction with the proteolytic machinery which releases the ARFs from inhibition, allowing activation of gene expression (Dharmasiri et al. 2004; Gray et al. 2000; Gray et al. 2001; Ward et al. 2001). Therefore, auxin actually controls the turnover of Aux/IAA proteins.

Considering the role of ARFs in various physiological and developmental processes and prospective utilization for better yielding crops, ARF gene families have been identified and characterized in important commercial crops such as *Arabidopsis thaliana*, rice (*Oryza sativa*), sorghum (*Sorghum bicolor*), maize (*Zea mays*), tomato (*Solanum lycopersicum*), and Chinese cabbage (*Brassica rapa*) (Haiyang et al. 2010; Song et al. 2014; Wang et al. 2007; Xing et al. 2011). Soybean (*Glycine max*) and common bean (*Phaseolus vulgaris*) are model species of fabaceae family and are considered important legumes for human consumption. Both plants are rendered as ideal species of respective genera to study symbiosis with nitrogen fixing bacteria. Furthermore, importance of these two species is broadly discussed in human and animal feed with particular focus on dietary proteins and edible oil. The availability of soybean and common bean genome sequences (Schmutz et al. 2010, 2014) has enabled us to carry out in silico

genome-wide comparative analyses of uncharacterized gene families. The phylogeny and structure of the ARF gene family has been explored in soybean (Ha et al. 2013). Fifty-one GmARFs were reported to be evolved because of genome duplications. In addition to expression analysis, expression profiling using qRT-PCR was provided. However, motif prediction, promoter prediction, and exon–intron structure was lacking in the report. Furthermore, its phylogeny with other related species, such as the common bean, is still undiscovered and poorly understood. Moreover, the auxin response mechanism of these species is still unclear in stress condition. Ha et al. (2013) show that different stress-responsive gene expression in different tissues at the same developmental stage or in the same tissue at different developmental stages under water stress. The expression analysis in normal conditions would be provided important initially information to assist in the characterized functions of the ARFs.

Here, we report genome wide identification of ARF family genes in soybean and common bean by comparing the structure of the encoded proteins with those of their putative homologous ARF genes in *Arabidopsis*. To investigate duplication events, soybean and common bean chromosome sequence information was applied to map ARF transcripts to their corresponding genetic loci on chromosomes. In addition, we analyzed the transcriptome atlas of ARF genes in different tissues under normal conditions and found notable differential expression between groups, which indicates broad functional divergence in this gene family. Finally, we compared and identified ARF gene family members in soybean and common bean. Our results deliver valuable information about the ARF gene family in soybean and common bean to facilitate functional analysis. This comparative in silico genomics knowledge of two companion species is an attempt to define important evolutionary relationships, time divergence, and genomic duplications.

Materials and methods

Sequence collection

Seventy-three protein sequences encoding *Arabidopsis* ARF domain downloaded from AtTFDB (<http://arabidopsis.med.ohio-state.edu/AtTFDB/>) were used in the initial protein queries to Basic Local Alignment Search Tool (BLAST) against the soybean genome database (assembly Wm82.a2.v1) and common bean genome database (assembly *P. vulgaris* v1.0) Phytozome v10.3 (<http://phytozome.jgi.doe.gov/pz/portal.html>) (Goodstein et al. 2012), using the BLASTP program at the score value of ≥ 100 and e value $\leq e^{-10}$. To exclude overlapping genes, all candidate

ARF genes were aligned using Clustal Omega (<http://www.ebi.ac.uk/Tools/msa/clustalo/>) (McWilliam et al. 2013). For each candidate sequence, data was collected from Phytozome (www.phytozome.net) of the location on chromosomes, genomic sequences, full coding sequences (CDS), and amino acid sequences. The Simple Modular Architecture Research Tool (SMART; <http://smart.embl-heidelberg.de/>) was used to confirm each predicted ARF member.

Sequence alignment and phylogenetic analysis

All conserved domain sequences of 55 ARF family proteins from *G. max*, 25 protein sequences from *P. vulgaris* and 23 protein sequences from *Arabidopsis* were investigated via multiple alignment analyses using the Clustal X v2.1 program (Thompson et al. 1997) with default parameters. The phylogenetic analyses for all complete ARF protein sequences were generated and viewed using the Molecular Evolutionary Genetics Analysis (MEGA) v7.0 (Kumar et al. 2016) by the neighbor-joining method. Bootstrap analysis was performed using 1000 replicates.

Gene structure and construction of chromosome map

The exon and intron structures of soybean and common bean ARF genes were extracted by comparing predicted CDS with their corresponding genomic sequences using gene structure display server (GSDS; <http://gsds.cbi.pku.edu.cn/>) software. SoyBase (<http://soybase.org/soyseq/>) and Phytozome by BlastN were used to define starting positions on chromosomal locations. The physical output was drawn by MapInspect software (http://www.plantbreeding.wur.nl/uk/software_mapinspect.html). The position of the centromere was retrieved from Soybase (<http://soybase.org/>) and Legume Information System (<http://legumeinfo.org/home>).

Analysis of conserved motif and subcellular localization

To examine the structural divergence among soybean ARF genes, motifs of paralogous ARF proteins were identified statistically using MEME analysis online (<http://meme.nbcr.net/meme/>) with default settings, except that the maximum number of motifs was set at 15, the width of motif was set from 6 to 200 and the site of each motif was set from 2 to 600. The subcellular localization of soybean and common bean ARF protein sequences were predicted by sub-cellular localization predictor (CELLO) v2.5 (<http://cello.life.nctu.edu.tw/>).

In silico expression and promoter analysis

Fifty-five GmARF genes were introduced into microarray data from Genevestigator software to detect the expression profile in various tissues and under multiple environmental stimuli. Different predicted expression levels of all GmARF were obtained by heat map analyses. In common bean, RNA-Seq data of 25 PvARF were downloaded from the PvGEA Atlas online database (<http://plantgrn.noble.org/PvGEA/>). The expression profile of PvARF were determined in young flowers (PVFY), leaf tissue collected 5 days (PVL5), 21 days (PVLE) after plants were inoculated with and without effective rhizobium (PVLI) and also from fertilized plants collected at the same time of LE and LI (PVLFI), nodule tissue collected 5 days (PVN5) and 21 days with and without effecting (PVNE and PVNI), young pods (PVPY), pods approximately 9 cm long (PVPH), between 10 and 11 cm long (PVP1) and between 12 and 13 cm long, associated with stage 2 seeds (PVP2), whole roots separated from fix- nodules collected 5 days (PVR5), 21 days after with and without inoculation (PVRE and PVRI respectively), and at the same time as RE and RI (PVRF), root tips (PVRT), seeds are between 3 and 4 mm across and approximately 7 mg (PVSH), between 6 and 7 mm across and approximately 50 mg (PVS1) and between 8 and 10 mm across and between 140 and 150 mg (PVS2), whole root including root tips (PVYR) shoot tip (PVST), fully expanded 2nd trifoliate leaf tissue (PVYL) and all stem internodes above the cotyledon (PVYS). A heat map was generated using the HeatmapGenerator (Khomtchouk et al. 2014). The 1.5 kb upstream of the coding region were selected as the promoter sequence and downloaded from Soybase (soybean ARF genes) and Legume Information System (common bean ARF genes). These sequences were submitted to PlantCARE for in silico analysis to predict the *cis*-motifs regulated gene expression.

Analysis of ARF gene expansion patterns and dating the duplication events

Tandem and segmental duplications were traced to predict gene expansion analysis. Tandem duplicated clusters based on the gene locus were identified and reviewed using a maximum of ten extra genes between ARF family genes. To identify segmental duplications of each ARF gene in soybean and common bean and the cross-genome syntenic relationships between *Arabidopsis*, soybean, and common bean, genomes were searched in the PGDD database (<http://chibba.agtec.uga.edu/duplication/>). The anchors with synonymous substitution rate (*Ks*) values of <1.0 were collected in the range of 100 kb. The segmental duplication events were calculated with the following

formula: $T = Ks/2\lambda$, assuming clock-like rates (λ) of synonymous substitution of 6.1×10^{-9} substitutions/synonymous site/year (Proost et al. 2011).

Results

Identification and distribution of the ARF gene family in soybean

In present plant genomic research, *Arabidopsis* is a dicot model organism used to identify functional predicted genes in other organisms that share a close phylogenetic relationship, such as tomato, cucumber, citrus, soybean, and common bean. Moreover, there are at least 23 ARF family members, which have been extensively studied and reported to be involved in many physiological and biochemical processes in *Arabidopsis* (Donaldson et al. 2011; Okushima et al. 2005). With the aim of defining the soybean and common bean protein-containing ARF domains, we used the 23 known *Arabidopsis* ARF protein sequences (Table 1) as the initial sequences to BLAST the completed soybean and common bean genome sequences, searching for genes that encode proteins containing the ARF domain.

Table 1 ARF genes in *Arabidopsis*

Name	Generic name	Chromosome location	Locus ID
AtARF1		1	At1g59750
AtARF2	HSS	5	At5g62000
AtARF3	ETTIN	2	At2g33860
AtARF4		5	At5g60450
AtARF5	MP	1	At1g19850
AtARF6		1	At1g30330
AtARF7	NPH4	5	At5g20730
AtARF8		5	At5g37020
AtARF9		4	At4g23980
AtARF10		2	At2g28350
AtARF11		2	At2g46530
AtARF12		1	At1g34310
AtARF13		1	At1g34170
AtARF14		1	At1g35540
AtARF15		1	At1g35520
AtARF16		4	At4g30080
AtARF17		1	At1g77850
AtARF18		3	At3g61830
AtARF19		1	At1g19220
AtARF20		1	At1g35240
AtARF21		1	At1g34410
AtARF22		1	At1g34390
AtARF23		1	At1g43950

After manually removing overlapping genes, a total of 55 and 25 ARF genes were identified in the soybean genome and common bean genome, respectively (Online Resource 1, Table 2). The nomenclature system for GmARF and PvARF in the present study provisionally uses the names GmARF1 to GmARF55 and PvARF1 to PvARF25 to distinguish each of the ARF genes based on the homology between AtARF, GmARF and PvARF genes.

In addition, a total of 80 physical positions of candidate ARF genes were collected from the Phytozome database. Fifty-five GmARF were among all soybean chromosomes, from chromosome 1–20. Chromosome 13 had the highest density of ARF genes with 8 members, while only one ARF gene could be found in each of chromosomes 6, 9, and 20. Examination of each ARF gene location revealed that all GmARFs, excluding GmARF21, originated from segmental duplications or tandem duplications (Fig. 1). Twenty-five PvARF genes are distributed on 11 chromosomes in common bean (Fig. 2). Most of the genes were located on chromosomes 6 and 8 (4 genes each chromosome), with only one gene each on chromosome 1, 4, and 10. Among 25 PvARF genes, 14 were segmentally duplicated and no tandem duplications were determined. The subcellular localization of soybean and common bean proteins was also predicted using sub-cellular localization predictor (Online Resource 1). Most ARF genes were predicted to be nuclear localized, while three members of GmARF proteins and PvARF20 were predicted to be localized in the chloroplast.

Table 2 Number of ARF genes in soybean and common bean

	Number of ARF genes	
	Soybean	Common bean
Total genes	55	25
Group		
I	20	8
II	7	4
III	14	7
IV	12	6
V	2	0
Location		
Nuclear	52	24
Chloroplast	3	1
Domain		
DBD, ARF, CTD	43	20
DBD, ARF	12	5
No. exon		
<10	12	6
>10	43	19

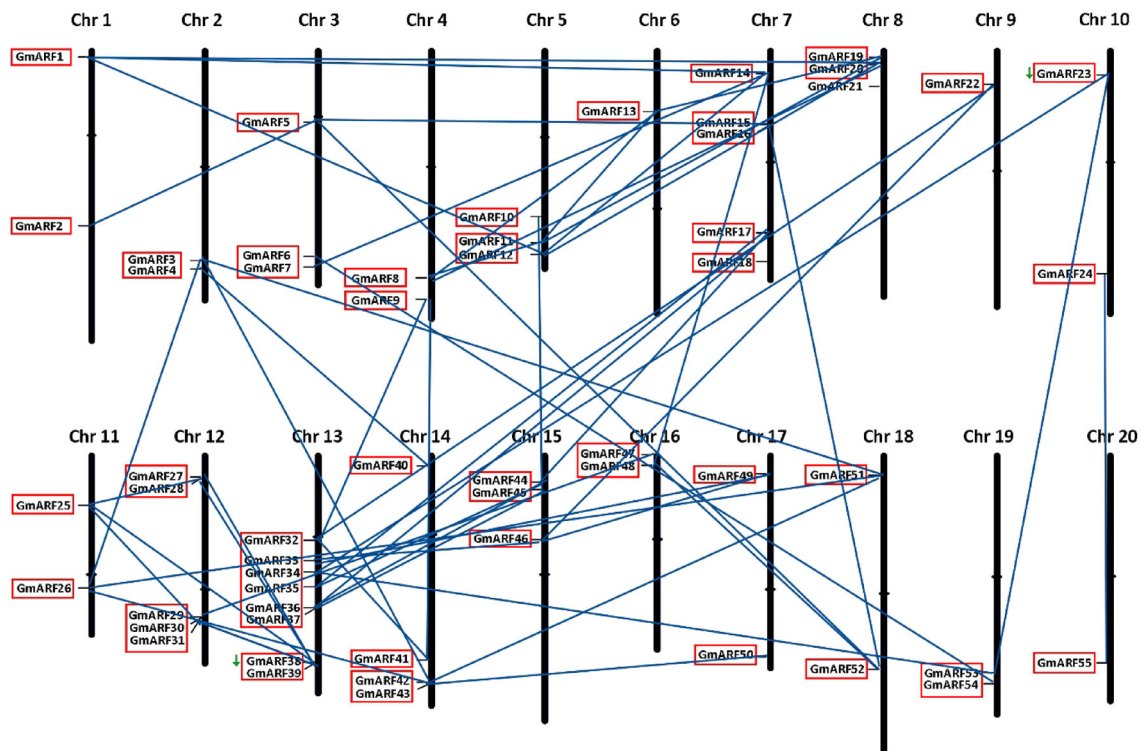


Fig. 1 Chromosome distribution of soybean (*G. max*) ARF genes. Segmentally duplicated genes are indicated by the red box while the tandem duplicated genes are indicated with vertical green lines.

Centromere positions are indicated in black and chromosome size is indicated by its relative length. Location and chromosome information was obtained from Phytozome

Multiple sequence alignment and structure analysis

An unrooted phylogenetic tree was constructed from alignments of the domain protein sequences of 103 ARF family members from *Arabidopsis* (23 genes), soybean (55 genes), and common bean (25 genes) to examine the evolutionary relationships of the ARF family between them. The phylogenetic distribution revealed that ARF genes group into five major classes, labeled class I, II, III, IV, and V, with well supported bootstrap values (Fig. 3). Thirty-three members were clustered in class I (twenty members from soybean to eight members from common bean), 13 members were clustered in class II (seven from soybean to four members from common bean), 26 members were clustered in class III (fourteen from soybean to seven from common bean), and 21 members were clustered in class IV (twelve from soybean to six from common bean). Notably, class V contained only two representatives in soybean (GmARF12 and GmARF55) and contained eight members from *Arabidopsis* (Fig. 3). Multiple alignment results indicated that most GmARF and PvARF proteins contained three characteristic regions including DBD, ARF, and CTD (Online Resource 1, 2). All ARF proteins had a highly conserved region of about 100 amino acid residues in their N-terminal portion corresponding to

the DNA-binding domain (DBD). A gene's middle region has been reported to function as an activation or repression domain (ARF) and as a C-terminal Aux/IAA domain (CTD) (Ulmasov et al. 1999a). To further investigate the characteristic regions of GmARF proteins, online MEME was employed to analyze motif distributions in 55 GmARF proteins and 25 PvARF proteins, with 15 individual motifs isolated. As predicted, most closely related members from the phylogenetic tree showed common motif compositions, suggesting functional similarities (Online Resource 3). Three domains of GmARF proteins were divided into 15 motifs. Three conserved motifs constituted the DBD domain. The ARF domain consisted of four motifs while the CTD domain corresponded to 3 motifs. Moreover, several specified unknown motifs were also identified in both soybean and common bean. To understand the structural components of ARF genes of soybean and common bean, a comparison of the full-length cDNA sequences with the corresponding genomic DNA sequences was obtained to examine the exon and intron organization. The CDS of the entire ARF family of both species were disrupted by introns, and the number of exons varied from two to fourteen (Online Resource 4). In general, the close homologs from the same group contain similar exon/intron structure with regard to intron number and exon length.

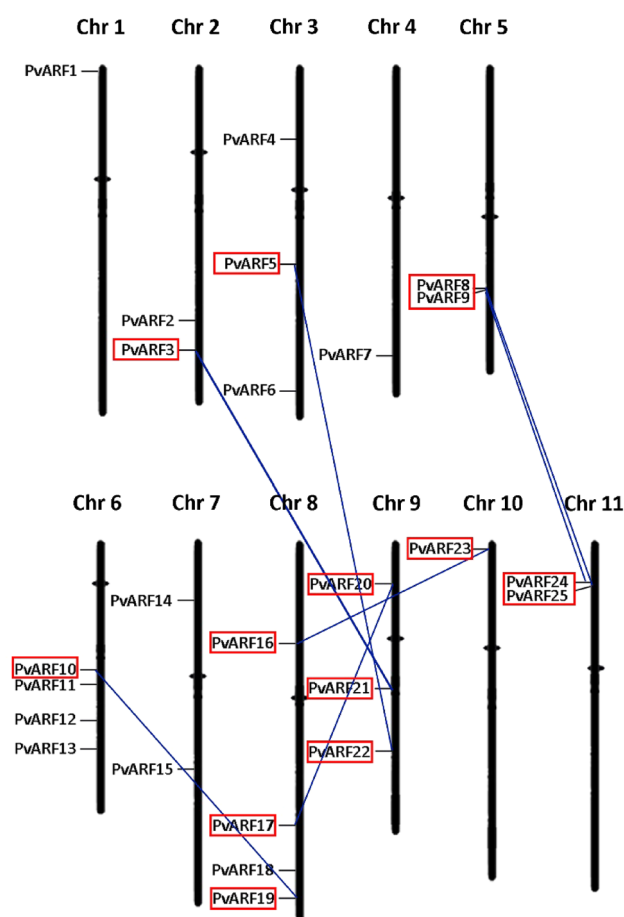


Fig. 2 Chromosome distribution of common bean (*P. vulgaris*) ARF genes. Segmentally duplicated genes are indicated by the red box. Centromere positions are indicated in black and chromosome size is indicated by its relative length. Location and chromosome information was obtained from Phytozome

Our results of intron/exon structure identification (Online Resource 4) showed that most of the DBD and ARF domain-containing protein genes in different groups or subgroups have very conservative exon numbers. In addition, other basic information of all ARF family members of both soybean and common bean, including domain number, group, CDS length, and gene length, are provided in Online Resource 1 and Table 2.

In silico expression and promoter analysis

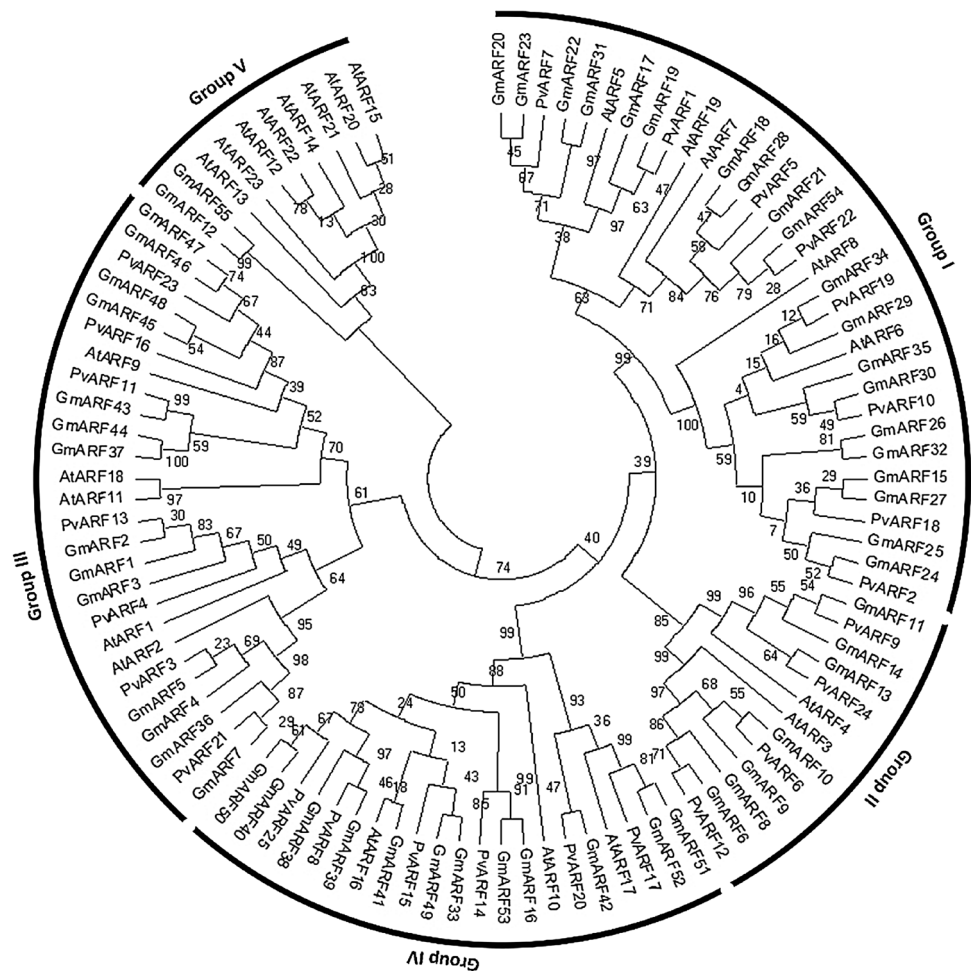
The previously completed transcriptome atlas of several plants made EST data available for comparison with the current study on soybean and common bean (Libault et al. 2010a; O'Rourke et al. 2014). We examined the expression patterns of soybean and common bean ARF genes across multiple tissues and organs to characterize and compare with the gene transcriptome atlas using developed RNA-

Seq web-based tools. Consequently, distinct transcript abundance patterns are readily identifiable in the data set for 55 GmARFs and 25 PvARFs, which displayed very broad expression spectra. However, sixteen GmARFs lacked expression information, including GmARF3, 4, 12, 17, 24, 25, 30, 31, 32, 39, 41, 43, 44, 49, 54, and 55. In soybean, we observed that accumulation of ARF gene transcripts was associated with various tissues, and expression patterns differed among each ARF gene member (Fig. 4). Of the analyzed ARFs, 55.3 % (31 of 38) were constitutively expressed in all of the five tested tissues, which suggests that GmARFs play a regulatory role during multiple developmental stages. By contrast, most GmARFs showed preferential expression in over 50 different tissues. RNA-Seq atlas data revealed that all GmARFs showed a high level of expression in root and shoot tissues, which suggests that GmARFs play an important role in regulating auxin synthesis. Approximately seven of these 38 (18.4 %) GmARFs showed the highest transcript accumulation level in root tissue, 10 (26.3 %) showed the highest transcript accumulation in shoot tissue, and surprisingly, none showed the highest transcript accumulation level in seed, leaf, and flower tissue. The wide expression of these genes (26 of 49 tested tissues) suggests that ARF genes from soybean are involved in the development of all organs or tissues under normal conditions.

In common bean, most genes have a broad expression spectrum (Fig. 5). Six genes (24 %) were found to show specific expression in shoot tissue. Interestingly, several ARF genes of common bean showed high expression in pod and nodule tissue.

PlantCARE, an available online computational tool, was helpful to determine the regulatory roles of ARFs under various stresses in the 1500 bp DNA sequence upstream of the translation initiation codon of ARF genes in soybean and common bean. In both soybean and common bean, four types of *cis*-acting elements were predicted to be significantly abundant in the ARF gene promoter regions, except in GmARF6 and PvARF12 owing to a lack of upstream sequence information (Online Resource 5). All ARF promoters of soybean and common bean possessed typical TATA and CAAT boxes; the core elements of promoters. There are 28 different types of light cycle related elements present in two species, such as GT1, G-box, Box II and Box 4, etc. Most of ARF promoters have at least one copy of Box 4. The mean number of copies (1.58) in the whole ARF gene family was found to be higher than other types of *cis*-acting elements except for TATA-box, CAAT-box, and unnamed-4. This result suggests that the Box 4, one part of the pal conserved DNA module array (pal-CMA1) involved with light responsiveness, tends to be enriched in the soybean and common bean ARF gene family. Twelve plant hormone responsive

Fig. 3 Phylogenetic tree of ARF genes among soybean, common bean, and *Arabidopsis*. Amino acid sequences of the ARF domains of soybean, common bean, and *Arabidopsis* were aligned with Clustal X. The phylogenetic tree was constructed using the neighbor likelihood method in MEGA 7.0



elements, such as AuxRE, AuxRE core, CGTCA-motif, ARE, ABRE, TCA-element, MBSI, MBSII, TATC-box, GARE-motif, TGA-box and TGA-elements constitute the second class. These observations suggest that soybean and common bean ARFs are likely to be significantly regulated by different types of hormones. Moreover, the third most abundant *cis*-acting element class consisted of elements that respond to external environmental stresses. W box, ERE, Box W1, TC-rich repeat, and HSE motif were found in most of ARFs, with responses to elicitation, fungal elicitors and heat stress, respectively. Skn-1 motif, required for high levels of endosperm expression, belong to the fourth type of *cis*-acting element that was abundantly found in the promoter regions of soybean and common bean ARF genes. PlantCARE identified 78.75 % (63 of 80) ARF genes containing Skn-1 elements. In addition, circadian, which involved in circadian control, also related to 67.5 % of ARF promoters. The presence of a diversity of *cis*-acting elements in the upstream regions of ARFs indicates that ARFs may function in different regulatory mechanisms in response to various stresses.

Expansion pattern of the ARF family

Duplicated genes provide the raw material for regeneration with new function gene. To understand gene duplication patterns in soybean and common bean, we generated expected values for tandem and segmental duplication. We searched for contiguous ARF genes in both the shared region and neighboring regions. Only two of the 55 genes found (3.6 %) in this family were tandem repeats in soybean (Fig. 1), indicating that tandem duplications contributed slightly to the expansion of this species. Similarly, one tandem duplication was detected in common bean. To compare gene duplication between soybean and common bean, the entirety of the ARF genes were searched for the following criteria: ≥ 80 % of the converted length of alignable sequence and ≥ 70 % similarity of the aligned region. In our results, 98.2–56.0 % of GmARF and PvARF genes, respectively, showed high conservation, indicating that soybean and common bean ARF families were formed through segmental duplication events (Table 3). We estimated the dates of the segmental duplication events by

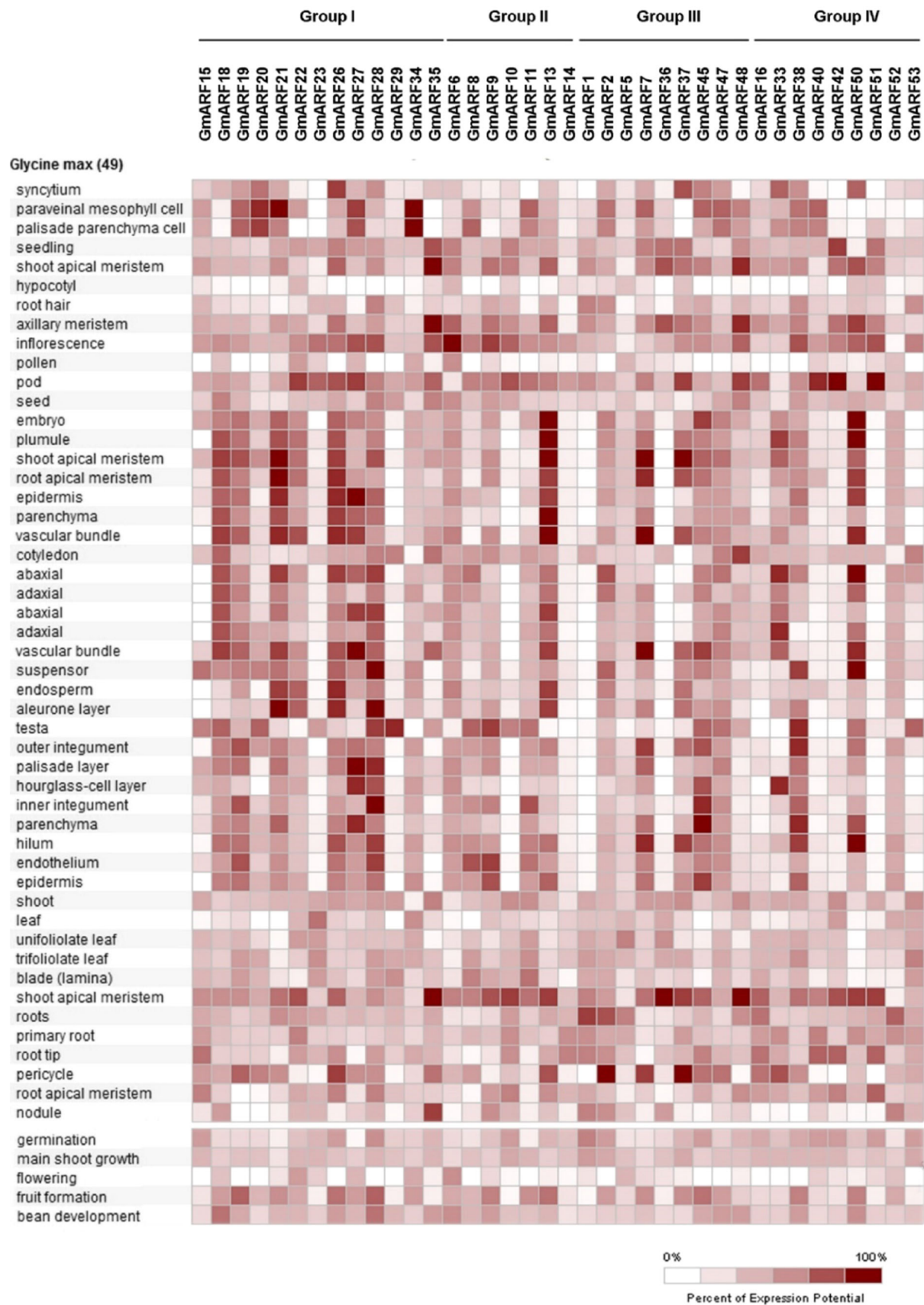


Fig. 4 Heatmap of 38 soybean ARF gene expressions. The highest and lowest expression levels are indicated in *red* and *white*, respectively. Seventeen ARF genes were excluded from the analysis due to the lack of expression in an organ or a period

using the K_s value, the synonymous substitution rate. The mean K_s values and the estimated dates for all segmental duplication events corresponding to ARF genes of soybean and common bean are listed in Table 3. In soybean, recent duplication events occurred during two periods, 10–20 mya (million years ago) and 40–60 mya, while most duplication events occurred only between 40 and 60 mya

in common bean (Schmutz et al. 2010). From these findings, we can hypothesize that segmental duplication is the main mechanism for expansion of the ARF gene family in both soybean and common bean, and that this mechanism is accompanied by tandem duplications and ARF gene duplications, more prevalently in soybean than in common bean. The results demonstrated that 24,975 syntenic blocks between soybean and *Arabidopsis* genome, 9785 blocks between common bean and *Arabidopsis* genome and 42,064 blocks between soybean and common bean were identified in this study. Moreover, 30 high confidence syntenic blocks include at least one pair of ARF

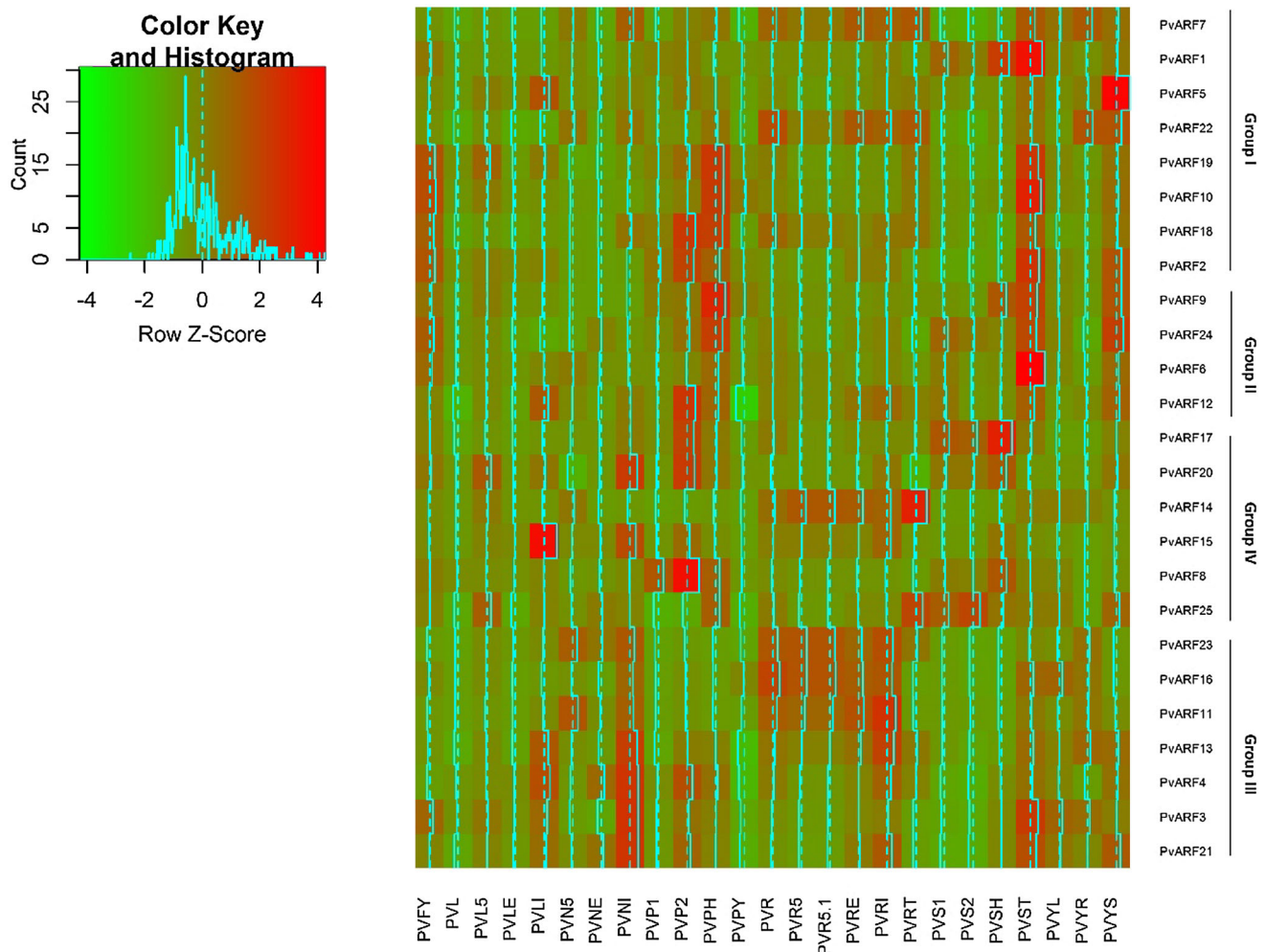


Fig. 5 Heatmap of 25 common bean ARF gene expressions. The highest and lowest expression levels are indicated in *red* and *white*, respectively. The expression profile of PvARF were determined in young flowers (PVFY), leaf tissue collected 5 days (PVL5), 21 days (PVLE) after plants were inoculated with and without effective rhizobium (PVLI) and also from fertilized plants collected at the same time of LE and LI (PVL5), nodule tissue collected 5 days (PVN5) and 21 days with and without effecting (PVNE and PVNI), young pods (PVPY), pods approximately 9 cm long (PVP3), between 10 and 11 cm long (PVP1) and between 12 and 13 cm long, associated with

stage two seeds (PVP2), whole roots separated from fix-nodules collected 5 days (PVR5), 21 days after with and without inoculation (PVRE and PVRI respectively), and at the same time as RE and RI (PVR5), root tips (PVRT), seeds are between 3 and 4 mm across and approximately 7 mg (PVSH), between 6 and 7 mm across and approximately 50 mg (PVS1) and between 8 and 10 mm across and between 140 and 150 mg (PVS2), whole root including root tips (PVYR) shoot tip (PVST), fully expanded 2nd trifoliate leaf tissue (PVYL) and all stem internodes above the cotyledon (PVYS)

Table 3 Estimates of the dates for the segmental duplication events of ARF family

Segment pairs	Number of anchors	<i>Ks</i> (mean $\pm$ SD)	Estimated time (mya)
GmARF1 & GmARF12	3	0.50 $\pm$ 0.14	41
GmARF11 & GmARF13	85	0.71 $\pm$ 0.11	58
GmARF14 & GmARF1	3	0.12 $\pm$ 0.04	10
GmARF14 & GmARF12	3	0.55 $\pm$ 0.14	45
GmARF14 & GmARF18	23	0.56 $\pm$ 0.14	46
GmARF16 & GmARF5	6	0.68 $\pm$ 0.13	56
GmARF16 & GmARF52	27	0.10 $\pm$ 0.04	8
GmARF17 & GmARF35	76	0.12 $\pm$ 0.04	10
GmARF19 & GmARF11	63	0.12 $\pm$ 0.02	10
GmARF19 & GmARF13	41	0.74 $\pm$ 0.11	61
GmARF20 & GmARF1	4	0.52 $\pm$ 0.14	43
GmARF20 & GmARF12	228	0.10 $\pm$ 0.04	8
GmARF22 & GmARF46	436	0.15 $\pm$ 0.02	12
GmARF23 & GmARF34	152	0.33 $\pm$ 0.06	27
GmARF23 & GmARF53	139	0.63 $\pm$ 0.16	52
GmARF25 & GmARF28	9	0.15 $\pm$ 0.03	12
GmARF27 & GmARF39	11	0.59 $\pm$ 0.15	48
GmARF3 & GmARF26	14	0.48 $\pm$ 0.06	39
GmARF3 & GmARF42	203	0.11 $\pm$ 0.01	9
GmARF3 & GmARF51	60	0.48 $\pm$ 0.06	39
GmARF30 & GmARF39	2	0.09 $\pm$ 0.03	7
GmARF31 & GmARF25	6	0.75 $\pm$ 0.12	61
GmARF31 & GmARF28	4	0.68 $\pm$ 0.12	56
GmARF32 & GmARF22	70	0.48 $\pm$ 0.14	39
GmARF32 & GmARF41	19	0.14 $\pm$ 0.04	11
GmARF33 & GmARF46	75	0.48 $\pm$ 0.14	39
GmARF33 & GmARF49	175	0.12 $\pm$ 0.04	10
GmARF36 & GmARF45	132	0.12 $\pm$ 0.02	10
GmARF37 & GmARF17	9	0.60 $\pm$ 0.15	49
GmARF37 & GmARF35	8	0.64 $\pm$ 0.15	52
GmARF37 & GmARF44	222	0.10 $\pm$ 0.04	8
GmARF38 & GmARF25	5	0.88 $\pm$ 0.17	72
GmARF38 & GmARF28	5	0.79 $\pm$ 0.16	65
GmARF38 & GmARF31	8	0.18 $\pm$ 0.03	15
GmARF40 & GmARF4	105	0.09 $\pm$ 0.02	7
GmARF42 & GmARF26	14	0.47 $\pm$ 0.06	39
GmARF44 & GmARF17	89	0.58 $\pm$ 0.16	48
GmARF44 & GmARF35	7	0.62 $\pm$ 0.17	51
GmARF45 & GmARF10	202	0.10 $\pm$ 0.04	8
GmARF46 & GmARF49	76	0.51 $\pm$ 0.09	42
GmARF47 & GmARF29	2	0.07 $\pm$ 0.02	6
GmARF48 & GmARF14	140	0.12 $\pm$ 0.02	10
GmARF48 & GmARF8	17	0.56 $\pm$ 0.17	46
GmARF5 & GmARF2	8	0.19 $\pm$ 0.02	16
GmARF5 & GmARF52	7	0.64 $\pm$ 0.12	52
GmARF50 & GmARF43	14	0.16 $\pm$ 0.02	13
GmARF51 & GmARF26	39	0.13 $\pm$ 0.01	11
GmARF51 & GmARF42	57	0.46 $\pm$ 0.06	38
GmARF52 & GmARF48	17	0.56 $\pm$ 0.08	46
GmARF53 & GmARF34	90	0.97 $\pm$ 0.15	80

Table 3 continued

Segment pairs	Number of anchors	Ks (mean $\pm$ SD)	Estimated time (mya)
GmARF55 & GmARF24	331	0.13 $\pm$ 0.03	11
GmARF6 & GmARF54	623	0.19 $\pm$ 0.02	16
GmARF7 & GmARF14	19	0.62 $\pm$ 0.12	51
GmARF8 & GmARF11	19	0.70 $\pm$ 0.11	57
GmARF8 & GmARF13	282	0.09 $\pm$ 0.02	7
GmARF8 & GmARF19	21	0.71 $\pm$ 0.11	58
GmARF9 & GmARF32	23	0.80 $\pm$ 0.09	66
GmARF9 & GmARF41	10	0.67 $\pm$ 0.34	55
PvARF3 & PvARF21	41	0.83 $\pm$ 0.12	68
PvARF5 & PvARF22	20	0.59 $\pm$ 0.11	48
PvARF8 & PvARF25	125	0.71 $\pm$ 0.15	58
PvARF9 & PvARF24	4	0.50 $\pm$ 0.10	41
PvARF10 & PvARF19	33	0.47 $\pm$ 0.06	38
PvARF16 & PvARF23	6	0.63 $\pm$ 0.14	52
PvARF17 & PvARF20	4	0.82 $\pm$ 0.34	67

orthologous genes between soybean and *Arabidopsis*, accounting for about 54 % of the GmARFs, while only seven syntenic blocks (28 % of the PvARFs) were found between common bean and *Arabidopsis* (Online Resource 6).

Discussion

The ARF gene family encodes transcription factors that play a central role during plant growth and developmental processes (Guilfoyle et al. 2007). Therefore, to better elucidate the function of soybean and common bean ARFs in affecting specific auxin responses, the present study portrays the main structural features of the ARF gene family in soybean and common bean. To isolate the complete array of ARF family members and perform expression profiling of these transcriptional regulators, we used the recent sequencing results of the soybean genome and transcriptome profiles to identify 55 GmARFs and 25 PvARFs (Online Resource 1). In soybean, there were much more ARF members than ARF genes encoded in sweet orange (13), *Arabidopsis* (23), rice (25) and poplar (39) (Hagen et al. 2002; Kalluri et al. 2007; Li et al. 2015; Shen et al. 2010). The first reason may be that the soybean genome is many times larger in size than those of *Arabidopsis* or rice (1100, 450, and 130 Mb, respectively). The large-scale duplication events that occurred in the evolution of soybean may be the second reason for the higher number of ARFs seen in this genome in comparison with sweet orange or common bean (Li et al. 2015).

A phylogenetic tree was applied on a well-distributed set of 55 and 25 ARF genes in soybean and common bean, clearly demonstrating that these genes can be divided into

five groups. Our classification was further supported by the results of motif structure, exon/intron examination, and pattern expression. Phylogenetic analyses could be used to predict protein functions between different species (Li et al. 2015; Shen 2013). Therefore, combining soybean, common bean, and *Arabidopsis* ARF proteins to construct phylogenetic tree would not only determine the relationship of ARF proteins, but would also help to predict the functions of the soybean and common bean ARF proteins based on the functions of the *Arabidopsis* ARF proteins found in the same groups (Fig. 3). For example, AtARF5, 6, 7, 8, 19 were found to be regulators in auxin mediated morphogenesis, flower development, or lateral root formation and gravitropism in *Arabidopsis*, which presently grouped with most soybean ARFs genes and 8 PvARFs in group I (Fukaki et al. 2006; Harper et al. 2000; Krogan et al. 2012). Seven members of GmARFs and four members from PvARFs with *Arabidopsis* AtARF3 and AtARF4 grouped together into group II, which regulates floral organs, developmental timing, and patterning in *Arabidopsis* (Fahlgren et al. 2007; Finet et al. 2010). Similarly, 14 and 7 of soybean and common bean ARF genes, respectively, were assembled together with AtARF1, 2, 9, 11, and 18 in group III, which regulates the leaf senescence process, floral organ abscission, or auxin homeostasis (Ellis et al. 2005; Guilfoyle et al. 2007). The group IV contained 12 GmARFs, 6 PvARFs, and AtARF10, 16, and 17 which were shown to affect root patterning, gravitropism, and auxin sensitivity from germination to post-germination stages under the regulation of the microRNA160 family (Blilou et al. 2005; Liu et al. 2009). Interestingly, group V only contained two members from soybean, GmARF12 and GmARF55. Additionally, the pattern and number of modular or portable domains present in the ARF

sequences also can be used to predict function (Guilfoyle et al. 2007). In general, ARF proteins share three characteristic regions: the B3-type DNA binding domain in the N terminal (BDB), an activation or repression domain (ARF) in a middle region, and the dimerization domain of the C terminal (CTD) (Hagen et al. 2002). In this study, protein sequence alignment of the GmARF and PvARF with their *Arabidopsis* counterparts confirmed that all proteins had a typical ARF-type structure (Online Resource 1). In a comparison with the ARF members identified in other plants, CTD-truncated in soybean (23.63 %) is higher than in sweet orange (21.05 %), *B. rapa* (22.58 %), and also common bean (20 %) (Mun et al. 2012; Wu et al. 2011).

Furthermore, the promoter analysis showed differential expression patterns of ARF in the soybean and common bean families. Four auxin responsive elements were identified in promoter of both species, especially TGA element (AACGAC) with 25 % (20 of 80 ARF genes). Moreover, the elements were distributed across three main functional categories, including light signaling perception, biotic and abiotic stresses, and developmental processes. Many GmARF and PvARF genes containing many light responsive elements were found in this study, it will provide more evidence of the interplay between auxin and light cycle pathway. Surprisingly, Skn-1 motif, which is required for endosperm-specific expression, was found to be abundant in all groups. This result may provide an explanation for the increased expression of ARF proteins during late embryogenesis. Moreover, the promoter analysis appears to support the expression analysis, in which most genes showed the highest transcript accumulation level in tissue and development stage such as root, shoot tissue, germination and bean formation (Figs. 4, 5). Since the function of *cis*-acting elements is to regulate gene expression, we speculated that the reason for this phenomenon might be due to the variety and high detection of several motifs in the predicted promoter region. The variation in the number and location of these motifs may affect the different expression patterns and also related with the key role of ARF protein in plants.

Recently, the transcriptome atlas of soybean genes was reported in eight tissues (nodules, SAM, flower, green pods, roots, root hairs, root tips, and leaves) (Ha et al. 2013; Libault et al. 2010b). Thus, we utilized 38 GmARF genes to show organ/tissue-specific expression patterns in 51 soybean tissues at five development stages. While 25 PvARF were present in 25 tissues, though not in root hair, the expression patterns of ARF genes suggested that the encoded proteins may perform both unique and redundant functions. In previous studies, some *Arabidopsis* ARF genes were expressed during flower development (Pekker et al. 2005). However, both the GmARFs and PvARFs showed low expression in the flowering stages in our study.

This was supported by a previous study by Li et al. (2015) on woody plants (sweet orange), which supported the hypothesis that expression is reduced during bloom. Although the roles of ARF genes in these processes are not yet known, given the similarity of soybean and common bean ARF genes to *Arabidopsis* ARF genes, the possibility of ARF genes participating in these different developmental processes in soybean and common bean cannot be excluded.

Previous studies concluded that at least two rounds of duplications probably occurred in the *Arabidopsis*, common bean, and soybean genomes, with many losses and rearrangements leaving a mosaic of “segmental duplications” or “duplication blocks” (Proost et al. 2011; Shoemaker et al. 2006). Most duplication blocks appear to have originated from one round of polyploidy that probably occurred 10–50 mya, before the evolution of the genus Brassica but after the separation of Brassicaceae from other closely related dicot families (Simillion et al. 2002). Several members of a specific gene family of a particular organism are the natural products generated from the organism’s long evolutionary history. The number of members of a gene family reflects a succession of genomic rearrangements and expansions due to extensive duplication and diversification that frequently occurred over the course of evolution. The whole genome and segmental duplication events are logically the contributors of increasing ARF gene numbers in soybean, rice, and *Arabidopsis* after their divergence. Based on the phylogenetic analysis in this study, no sister pairs of PvARF were identified, while 13 sister pairs of GmARFs were formed but only one of these pairs (GmARF51-52) were genetically linked to each other as compared with their corresponding chromosomal locations. On the other hand, all closely linked GmARF loci such as GmARF 15 and 16 on chromosome 7; GmARF 19 and 20 on chromosome 8; GmARF 29, 30, and 31 on chromosome 12; and GmARF36, 37, 38, and 39 on chromosome 13 are not paired together into the same sister groups. Results of the present study showed only two apparent tandem duplication events. Furthermore, most pairs of segmentally duplicated genes in soybean and 19 pairs in common bean focus on the period of 40–60 mya. The long evolutionary period has provided many opportunities for functional divergence in the genes that arose from duplications. Our results did not reveal evidence that other pairs of ARF genes in soybean and common bean originated from duplicated blocks. Therefore, most of the ARF genes may have lost their paralogous genes after genome duplication (Simillion et al. 2002). All of the evidence suggests that the evolutionary patterns of the ARF gene family differ between soybean, common bean, and other higher plants.

The results provided comprehensive information on the ARF families of both soybean and common bean, including gene structures, chromosome distribution, phylogenetic relationships, promoter analysis, expression patterns, and duplication events. The members of the ARF family are differentiated in soybean and common bean due to highly different genome length. Duplication events between the two species are close homologs. Studies of chromosomal distribution, phylogenetic, segmental, and tandem duplication of the ARF genes have provided valuable insights on the evolutionary aspects of the soybean and common bean genomes. We hypothesize that segmental duplications may be the dominant mechanism of gene amplification to expand the ARF family in both soybean and common bean. The large scale expression profiling of GmARF and PvARF allowed us to understand the functions and potentially the ARF transcription factors in an important model system.

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Compliance with ethical standards

Conflict of Interest Bao Le, Muhammad Anjad Nawaz, Hafiz Mamoon Rehman, Thu Le, Seung Hwan Yang, Kirill S. Golokhvast, Eunho Son and Gyuha Chung declares that they have no conflict of interest.

Research involving human and animal participants This article does not contain any studies with human subjects or animals performed by any of the authors.

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